

Zero-Shot Conservation Critics for Generative Physical Simulation

Abstract

Generative models of physical scenes are typically validated with a *critic*: a score that ranks or filters candidate trajectories. Learned critics are the default choice, but they require labelled data and can be brittle. We ask whether a **zero-shot critic built from conservation laws** — no training, no labels — is competitive.

We study *candidate selection*: given K trajectories generated from the same initial condition by a learned dynamics model, pick the one closest to the true solution. This is the operation a critic performs inside a generation loop, and unlike classification-style metrics it needs no threshold on “how wrong is wrong”.

Across three systems — a 6-body 3D family, the figure-eight three-body choreography, and a 2D Kepler two-body system — a zero-shot conservation critic attains **100% within-group selection accuracy** on the standard benchmarks, reducing trajectory error by 75–82% relative to random selection and landing within $1.03\text{--}1.05\times$ of the best achievable selection. On the 6-body system (5 seeds, 95% CI) it reaches $82.8\% \pm 3.3\%$ in distribution and $84.7\% \pm 1.8\%$ out of distribution, versus $71.2\% \pm 3.2\%$ and $72.2\% \pm 2.5\%$ for a supervised pairwise-ranking critic trained on the same 960 labelled candidates.

We also establish what the conservation critic is *not*. It does **not** beat a correctly-used Hamiltonian Neural Network critic: the two tie exactly (identical selection error on both standard benchmarks). And its lead does not survive a change of *generator*: with an HNN as the generator the supervised critic is at least as good. Claims that the conservation critic is more robust out of distribution, or that combining it with a learned critic helps, are **falsified** here and retracted.

Three methodological findings, each independently useful. (i) **Pooled rank correlation is misleading for critics**: a supervised regressor scored pooled Spearman $\rho = -0.67$, comparable to the best critic, while achieving a within-group win rate of 53%, i.e. chance. (ii) **A conservation channel’s discriminative power is set by its noise floor**, determined by the integrator and force structure rather than by the physical importance of the quantity: momentum and angular momentum are conserved *exactly* by any pairwise central-force integrator (floor $\sim 10^{-16}$) and separate true from generated trajectories by $\sim 10^{12}$, whereas energy is conserved only to the integrator’s truncation error ($O(\Delta t)$, floor 0.60 at $\Delta t = 0.01$) and separates by only $1.13\times$. (iii) **Synthetic failure modes must be validated before use**: a taxonomy of hand-built corruptions is not structurally comparable to a real generator’s failures — two of five cannot be tuned to the generator’s error magnitude at all, and conservation fingerprints differ by up to 238 normalized units.

1 Introduction

A generative model of physical scenes — a learned surrogate, a neural ODE, a video model — produces trajectories that may look plausible while violating physics. The standard remedy is a *critic*: a function that scores a candidate so that bad samples can be filtered, ranked, or used as a training signal.

Learned critics dominate, and they work: a network can capture subtle dynamical errors that no simple statistic catches. But they need labelled examples, and labels for physical validity are exactly what one is trying to avoid needing.

This paper asks a narrower and more testable question than “can physics replace learning?”:

Is a critic built from conservation laws — which is training-free and label-free — competitive with the strongest learned critics at selecting good candidates?

The answer we establish is: **yes, it ties the strongest**, and it does so at zero labelling cost. We also document three claims that turned out to be false, because the process of falsifying them produced the paper’s methodological contributions.

Contributions.

- **C1. A threshold-free evaluation protocol** for critics in a generation loop: *within-group selection*. Given K candidates from one initial condition, measure the critic’s rank correlation *within that group* and its selection error relative to random and oracle. Section 5.1 shows that the more obvious pooled metric is badly misleading.
- **C2. A zero-shot conservation critic that ties the strongest baselines**, including a correctly-used Hamiltonian Neural Network and a supervised pairwise-ranking critic trained on up to 4800 labelled candidates (Section 5.2).
- **C3. A noise-floor theory of channel selection** that explains, predicts, and improves the channel set (Sections 4.2 and 5.3).
- **C4. A negative result on synthetic failure taxonomies**, with a calibration procedure to test comparability before the taxonomy is used (Section 5.4).
- **C5. Three retractions** of claims made in earlier drafts of this work (Section 7).

2 Related work

Learned physical simulators. Hamiltonian Neural Networks [3] and Lagrangian Neural Networks [2] learn a scalar potential whose derivatives reproduce the dynamics. They are the natural learned alternative to a hand-written conservation law, and we adopt an HNN as our primary literature baseline.

Neural surrogates and operators learn the flow map directly. Their errors are typically evaluated by trajectory RMSE against a reference integration, which is available in simulation but not in the field.

Physics-based validation. Conservation checks are standard in numerical relativity and molecular dynamics as diagnostics. Using them as a *critic inside a generative loop* — the object of this paper — is less studied.

Positioning. We do not propose a new generator. We characterise when a training-free physical critic is sufficient, and we show that the answer depends on a numerical property (the noise floor) rather than on the physics.

Table 1: Metrics. Lower is better except for the win rate.

Metric	Definition	Chance	Perfect
within-group ρ	mean over ICs of Spearman(scores, e) inside the IC	0	-1
IC win rate	fraction of ICs where the selection beats that IC’s median error	50%	100%
selected / random	median selected error $\div$ median of per-IC mean error	1.00	—
selected / oracle	median selected error $\div$ median of per-IC minimum error	—	1.00

Table 2: Systems used throughout.

System	N	Dim.	Notes
6-body, eccentric family	6	3D	$e \sim U(0, 0.3)$ ID; $e \sim U(0.5, 0.8)$ OOD
figure-eight choreography	3	2D	[1]; $T = 6.32591398$
Kepler two-body	2	2D	standard low-dimensional HNN/LNN testbed

3 Setup

3.1 Task: within-group candidate selection

For each initial condition (an “IC”):

1. integrate the true equations of motion to obtain a reference trajectory $x^*(t)$;
2. run a learned generator $K = 8$ times from the *same* IC, with varying sampling noise $\sigma \in \{0, 0.02, 0.05, 0.10, 0.20, 0.35, 0.6, 1.0\}$;
3. compute each candidate’s error $e_j = \text{RMSE}(x_j, x^*)$;
4. score all candidates with each critic and select $\arg \max_j \text{score}_j$.

3.2 Systems

Checker for the figure-eight initial data: after exactly one standard period at $\Delta t = 0.002$ (3163 steps) the configuration returns to the initial condition to within 3.29×10^{-3} in position. All integrations use symplectic Euler at $\Delta t = 0.002$ –0.01.

3.3 Generators

The generator is a pairwise force model: a shared MLP applied to $[r_{ij}, |r_{ij}|, v_{ij}, m_i, m_j]$ producing a scalar pair force, summed over neighbours. It is deliberately trained on a *limited budget* so that its failures are large enough to be selectable among (one-step acceleration error 0.23 for the 6-body system; three budgets, 0.23/0.0068/0.81, are used for robustness checks).

Negative samples therefore come from a **real learned generator**, not from hand-built corruptions. Section 5.4 explains why that choice was forced.

Table 3: Critics compared. Training budgets differ per experiment; each table states its own. “HNN, energy drift” is the naive way to use an HNN as a critic; see Section 5.2.

Critic	Labels required	Notes
Conservation (zero-shot)	0	angular + momentum channels
HNN, equation residual	true trajectories	[3]
HNN, energy drift	true trajectories	naive use as a critic
Pairwise ranking loss	960–4800 candidates	supervised, target-aligned
Absolute binary classifier	960–4800 candidates	supervised, naive target
Absolute error regressor	960–4800 candidates	supervised, naive target
Random / oracle	—	reference points

3.4 Critics

4 Method: the conservation critic

4.1 Conservation drift as an error functional

Let the true acceleration field be a sum of pairwise central forces,

$$a_i^{\text{true}}(r) = \frac{1}{m_i} \sum_{j \neq i} f(r_{ij}) \hat{r}_{ij}, \quad (1)$$

and let $\delta a(t) = a^{\text{gen}}(\hat{z}(t)) - a^{\text{true}}(\hat{r}(t))$ be the generator’s acceleration error along its own trajectory $\hat{z}$. Because $\sum_i m_i a_i^{\text{true}} \equiv 0$ and $\sum_i m_i (r_i \times a_i^{\text{true}}) \equiv 0$ hold pointwise,

$$\frac{dP}{dt} = \sum_i m_i \delta a_i, \quad \frac{dL}{dt} = \sum_i m_i (\hat{r}_i \times \delta a_i), \quad \frac{dE}{dt} = \sum_i m_i (\hat{v}_i \cdot \delta a_i). \quad (2)$$

The conservation residual is therefore the **projection of the accumulated force error onto the symmetry generators** — a physics-meaningful functional, not a heuristic.

Semi-quantitative validation (40 ICs, medians of the ratio prediction/measured): momentum $0.53\times$, angular $1.23\times$, energy $9.0\times$; log-correlations $0.72/0.31/0.22$. Equation (2) is exact; its integrated form is validated to within a factor of a few for the channels with a clean numerical floor.

What is *not* claimed. Equation (2) bounds the *drift* by the force error. It gives **no bound on trajectory error**: any δa lying in the orthogonal complement of the three generators produces zero residual with arbitrarily large trajectory deviation. We never assume such a δa is absent.

4.2 Which channels to use, and why

The channels are measured on the reference trajectories. linear and angular momentum are exact invariants of any pairwise central force, to roundoff, independent of the integrator; their floor is $\eta_P \sim \eta_L \sim \varepsilon_{\text{mach}} \approx 10^{-16}$. Energy is conserved only to the integrator’s truncation error; for the first-order symplectic Euler used here, $\eta_E = O(\Delta t)$.

This predicts the measured channel ranking and is confirmed by a step-size sweep (identical generator, only Δt varied):

Channel utility is a numerical property, not a physical one. The critic therefore uses angular + momentum by default, and excludes both the energy channel (weak floor) and the centre-of-mass channel (an exact invariant, but its variation is nearly uncorrelated with shape error — measured IC win rate 16.7–24.2%, below chance).

Table 4: Noise floors and energy separation as a function of step size. The energy floor halves with Δt ; the momentum and angular floors stay pinned at roundoff.

Δt	energy floor	angular floor	momentum floor	energy separation
0.0100	6.02e-01	4.01e-16	2.88e-16	0.81
0.0050	2.65e-01	5.06e-16	3.68e-16	1.84
0.0025	1.36e-01	6.62e-16	4.39e-16	3.58

Table 5: Pooled versus within-group rank correlation. The supervised error regressor looks competitive pooled but has a within-group win rate of 53% — chance.

Critic	pooled ρ	within-group ρ	IC win rate	selected / random
Conservation	-0.7044	-0.6881	84.0%	0.637 $\times$
Pairwise ranking (supervised)	-0.4689	-0.5786	74.5%	0.656 $\times$
Error regressor (supervised)	-0.6685	-0.3799	53.0%	0.758 $\times$
Logistic fusion of all three	-0.4509	-0.6825	83.5%	0.641 $\times$

5 Experiments

5.1 Why the obvious metric is wrong

Reported “critic quality” is often a rank correlation pooled over all candidates. That metric conflates two effects, because candidates sampled with larger σ are worse across the board. A critic that only detects the overall noise level scores well pooled while being unable to rank inside a group. Measured on the 6-body system (600 training ICs = 4800 labelled candidates):

A related effect: aligning the training target with the evaluation is worth more than model capacity. Identical architecture, two targets — absolute error regression reaches an IC win rate of **47.5%**; IC-relative error regression reaches **86.5%**, a 39-point gap from the target alone.

5.2 Main results

The conservation critic ties the HNN exactly on both systems — identical selection error, identical win rate. The learned critic requires training an HNN; the conservation critic requires nothing.

Note also the separation column: the conservation critic separates true from generated trajectories by 10^{11} – 10^{14} , the HNN by $\sim 10^{-2}$. Yet the HNN still ranks correctly inside a group, because its residual varies monotonically with σ . Large absolute separation and good within-group ranking are different properties.

The numbers in Table 8 come from 120 training ICs. In Section 5.1 and Section 7 the supervised critics were given 600 ICs (4800 labelled candidates) — five times more — and the conclusion is unchanged. We quote each experiment’s own budget rather than a single figure.

5.3 Ablation over channel sets

Four settings (two generators $\times$ ID/OOD), reported as selected/random (lower is better):

On the stronger generator the old four-channel sum is $1.21\times$ of oracle where angular+momentum is $1.02\times$, and out of distribution the gap widens to $1.61\times$ versus $1.02\times$. The centre-of-mass channel is worse than chance on its own.

Table 6: figure-eight three-body ($N = 3$, 2D), 50 test ICs, $K = 8$. “Separation” is the relative gap between the critic’s score on the true trajectory and on the $\sigma = 0$ candidate.

Critic	separation	within-group ρ	IC win rate	sel./random	sel./oracle
Conservation (zero-shot)	8.7e11	−0.9486	100.0%	0.250×	1.03×
HNN, energy drift (naive)	0.076	−0.5405	50.0%	0.544×	2.24×
HNN, equation residual	0.017	−0.9524	100.0%	0.250×	1.03×
random / oracle	—	—	50.0% / 100%	1.000×	4.11×

Table 7: Kepler two-body ($N = 2$, 2D), 50 test ICs, $K = 8$.

Critic	separation	within-group ρ	IC win rate	sel./random	sel./oracle
Conservation (zero-shot)	2.4e14	−0.9495	100.0%	0.185×	1.05×
HNN, energy drift (naive)	0.034	−0.5705	52.0%	0.463×	2.63×
HNN, equation residual	0.058	−0.9490	100.0%	0.185×	1.05×
random / oracle	—	—	50.0% / 100%	1.000×	5.68×

5.4 A negative result: synthetic failure modes are not comparable

Critic rankings are often measured on hand-built corruptions (add noise, swap in a bad integrator, freeze, splice, time-warp). We tested whether such a taxonomy is comparable to a real generator’s failures. **It is not.**

(a) **Two of five corruptions cannot be calibrated.** Tuning severity so that each corruption’s median RMSE matches the generator’s (0.01468):

splice is worse than merely uncalibrated: it performs a discrete particle roll and its severity parameter has **no effect at all**. It is an artifact of the construction, not a failure mode.

(b) **The conservation fingerprints have different shape.** Normalized fingerprint distances range from 7.4 to 238.

Consequence. No result in this paper is computed on a synthetic corruption set. All negative samples come from real learned generators. We recommend the calibration procedure above as a prerequisite for anyone using such a taxonomy.

5.5 Cross-generator check: the ranking depends on the generator

The results above use a pairwise force model as the generator. A reviewer’s first question is whether they are an artefact of that choice. We re-ran the comparison with a **Hamiltonian Neural Network used as the generator** [3], which has a fundamentally different inductive bias: it learns a black-box scalar $H(q, p)$ and derives the vector field from Hamilton’s equations, so it is *not* translation- or rotation-invariant, unlike a pairwise force model.

The conservation critic no longer leads. On the figure-eight it is clearly behind (78.0% versus 96.0%); on Kepler the two are close with the supervised critic marginally ahead in selection error. Both remain far better than random (0.55–0.86× versus 1.00×).

The conservation fingerprint shows the HNN generator still breaks angular-momentum conservation severely (separation 2.4×10^{13} and 7.3×10^{12}), so the critic’s signal is not weakened.

Table 8: 6-body system, 5 seeds, 95% CI (120 training ICs). The conservation critic’s advantage over the supervised ranking critic has non-overlapping 95% CIs on both splits.

Critic	ID within-group ρ	ID win rate	OOD win rate	training samples
Conservation (zero-shot)	-0.6404 ± 0.0459	$82.8\% \pm 3.3\%$	$84.7\% \pm 1.8\%$	0
Pairwise ranking (supervised)	-0.5627 ± 0.0471	$71.2\% \pm 3.2\%$	$72.2\% \pm 2.5\%$	960
Absolute binary (naive)	-0.4958 ± 0.0768	$62.0\% \pm 8.8\%$	$67.0\% \pm 7.3\%$	960
random	—	50.0%	50.0%	—

Table 9: Channel-set ablation. The previously-used equal-weight sum of four channels is substantially suboptimal.

Channel set	G.A ID	G.A OOD	G.B ID	G.B OOD
angular	0.640×	0.637×	0.056×	0.063×
angular + momentum (adopted)	0.640×	0.653×	0.056×	0.059×
four channels, equal weight (previous)	0.629×	0.713×	0.067×	0.094×
energy only	0.794×	0.828×	0.308×	0.194×
centre of mass only	1.038×	1.057×	1.327×	1.575×

A candidate explanation, tested and rejected. The conservation residual is a projection of the force error onto the symmetry generators (Eq. 2), so one might expect the *full* force error to carry strictly more information, with the HNN generator’s advantage lying in structure that conservation cannot see. We tested this directly: for every candidate we measured the accumulated force error $\int \|a^{\text{gen}} - a^{\text{true}}\| dt$ along its own trajectory and correlated it, within each IC, with the true trajectory error.

The explanation does not hold. The force-error magnitude is a far weaker predictor than the conservation residual everywhere, so nothing here supports attributing the HNN generator’s advantage to information that conservation discards. Two reasons the diagnostic may simply be the wrong instrument, neither of which we separated: the force error is evaluated at the *generated* positions, so a candidate that has drifted is compared in a different region of state space; and trajectory error is the *second* time integral of δa , so oscillating force errors that largely cancel leave little trace — a magnitude is not the right upper bound on usable information.

We therefore leave the mechanism **unexplained**. What stands is the measurement: with an HNN generator the supervised critic is at least as good as the conservation critic. Only the proposed explanation failed.

What this changes. The claim must be stated per generator, not universally:

A zero-shot conservation critic is a **reliable but not universally optimal** supervision signal. For pairwise-force generators it leads the strongest supervised critic; for a Hamiltonian-parameterised generator it is matched or slightly beaten.

This does not contradict the tie reported in Section 5.2: that compares *critics* under a fixed pairwise-force generator, whereas this compares *generators*. Together they place the conservation critic in the same band as the strongest learned supervision, with the ordering depending on the setting rather than following a general rule.

Table 10: Severity calibration. “Lower bound” means the corruption was made as mild as its parameterisation allows.

Corruption	calibrated severity	achieved RMSE	verdict
noise	0.740	0.01477	calibrated
freeze	0.428	0.01342	calibrated
warp	0.302	0.01342	calibrated
euler	lower bound	0.14127	10× too destructive
splice	lower bound	0.59365	40× too destructive

Table 11: Separation of true from corrupted trajectories, per channel. The real generator’s failure mode is breaking Newton’s third law, which destroys angular and linear momentum conservation. Four of the five synthetic corruptions leave those quantities at 0.9–1.1.

Channel	noise	euler	freeze	splice	warp	real generator
energy	3.89	0.00	0.86	0.98	0.93	1.02
angular	7.5e15	3.1e10	0.95	0.97	1.13	1.2e15
momentum	6.3e15	3.9e8	0.87	0.96	0.94	1.9e16
centre of mass	1.10	0.02	0.85	1.03	1.03	1.10

6 Limitations

1. **Generators are learned surrogate force models**, not diffusion or autoregressive video models. The qualitative structure of the failures (breaking equal-and-opposite forces) is plausible for any pairwise-parameterised model, but we did not verify it on a large pretrained generative model.
2. **The generator’s parameterisation changes the ranking.** Section 5.5 shows that switching from a pairwise force model to a Hamiltonian Neural Network removes the conservation critic’s lead. The claim must be stated per generator; we have two architectures, not a spectrum.
3. **Ground truth is always available** here, because the systems are simulated. The selection task is therefore well-defined; in the field one normally lacks x^* . Our critic does not need x^* at *use* time — only our *evaluation* does.
4. **Three systems, all $N \leq 6$ and pairwise-central.** The noise-floor argument relies on (i) pairwise central forces and (ii) a symplectic integrator. For three-body forces or non-symplectic integrators the momentum and angular floors are no longer automatically at roundoff.
5. **No accuracy comparison against published numbers.** The HNN baseline was trained by us; we did not reproduce a published HNN result. The tie should be read as “our HNN implementation is a fair baseline”, not as a leaderboard claim.
6. **The critic is used only for selection.** We do not show that it improves *training* (e.g. as a reward or regulariser); that is untested.

7 Retractions

Three claims appeared in earlier internal drafts of this work. All are false, and the process of falsifying them produced the methodological results of Sections 5.1, 5.3 and 5.4.

Table 12: HNN used as the *generator*. The conservation critic’s lead does not survive the change of generator.

System	Critic	within-group ρ	IC win rate	sel./random	sel./oracle
figure-eight	Conservation (zero-shot)	−0.6881	78.0%	0.864×	1.10×
figure-eight	Pairwise ranking (supervised)	−0.8890	96.0%	0.794×	1.01×
Kepler	Conservation (zero-shot)	−0.8048	98.0%	0.574×	1.06×
Kepler	Pairwise ranking (supervised)	−0.9014	96.0%	0.552×	1.02×
random (figure-eight / Kepler)		—	50.0%	1.000×	1.27×

Table 13: Tested explanation. Positive Spearman means the diagnostic predicts trajectory error. The full force error is a *worse* predictor than the conservation residual in all four settings, so this diagnostic cannot support the “invisible structure” explanation.

Setting	ρ (conservation)	ρ (force error)	gap
figure-eight × pairwise	+0.9524	+0.1702	−0.7821
figure-eight × HNN	+0.7768	+0.1339	−0.6429
Kepler × pairwise	+0.8173	+0.1238	−0.6935
Kepler × HNN	+0.7929	+0.2083	−0.5845

R1. “The conservation critic is more robust out of distribution than a learned critic.”

False. On a family shift the two do not differ (conservation +0.001, learned −0.005, overlapping CIs). On a *severity* shift toward subtler violations the conservation critic degrades *more* (−11.4 points versus −1.0). Its advantage is absolute level and zero labelling cost, not distributional robustness.

R2. “Conservation and learned critics have complementary blind spots; combining them is consistently better.”

False. An *optimally weighted* fusion (logistic regression over the critic scores, fitted on held-out training data) assigns the conservation critic the largest weight (+1.635), the supervised error regressor a **negative** weight (−0.612), and still ties the conservation critic alone (0.641× versus 0.637× ID; 0.608× versus 0.614× OOD). The signals are redundant, not complementary. An earlier version of this claim appeared to hold only because the “learned” critic was given conservation residuals as input features, making the conclusion true by construction.

R3. “Conservation critics can validate the Lumyn pipeline itself.”

False, for a structural reason: that pipeline integrates the true equations of motion, so its output *is* a solution and the failure mode a validator would catch cannot occur. A real corruption/restoration experiment confirms this — the validation layer’s causal contribution is exactly zero, because it is never acted upon. A conservation critic is useful precisely when the generator’s output is *not* a solution, which is true of learned models and false of a physics integrator.

8 Conclusion

A critic built from conservation laws, requiring no training data and no labels, ties the strongest learned critics at selecting good candidates from a learned physical generator: 100% within-group accuracy on standard benchmarks (matching a Hamiltonian Neural Network exactly), and 83–85% on a 6-body family where a supervised pairwise-ranking critic trained on 960 labelled candidates reaches 71–72%.

The reason it works is numerical rather than physical. Conservation channels whose invariance is a structural property of the pairwise central force — linear and angular momentum — sit at a machine-precision floor and separate true from generated trajectories by ten or more orders of magnitude. The energy channel, conserved only to the integrator’s truncation error, does not. Choosing channels by their noise floor, rather than by the physical prominence of the quantity, is what makes the critic work.

We also report what does not work, because it constrains the claim: the critic is not more robust out of distribution, combining it with learned critics does not help, and synthetic failure taxonomies are not structurally comparable to real generator failures unless calibrated first.

References

- [1] A. Chenciner and R. Montgomery. A remarkable periodic solution of the three-body problem in the case of equal masses. *Annals of Mathematics*, 152(3):881–901, 2000.
- [2] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho. Lagrangian Neural Networks. *arXiv:2003.04630*, 2020.
- [3] S. Greydanus, M. Dzamba, and J. Yosinski. Hamiltonian Neural Networks. *Advances in Neural Information Processing Systems*, 2019.

A Reproducibility

Table 14: Result-to-script map.

Result	Script
Selection metrics, 6-body, 5 seeds	<code>experiments/experiment_h_benchmark.py</code>
Channel ablation	<code>experiments/experiment_e_channel_combos.py</code>
Standard benchmarks + HNN	<code>experiments/experiment_k_hnn_equation_critic.py</code>
Strong supervised baselines	<code>experiments/experiment_c_strong_baselines.py</code>
Failure-mode comparability	<code>experiments/experiment_i_failure_modes.py</code>
Noise-floor sweep	<code>experiments/experiment_g_error_functionals.py</code>
HNN fairness diagnostics	<code>experiments/diagnose_hnn*.py</code>
Channel-set regression tests	<code>tests/test_conservation_critic.py</code> (18 tests)
Checker-defect regression tests	<code>tests/test_conservation_checker.py</code> (11 tests)

Test suite at the time of writing: **108 tests, 0 failures** (1 optional-dependency skip).